Project Report

Title: Mood-Based Music Player using Python

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1. Project Overview

This project is a simple Mood-Based Music Player developed using Python. It allows the user to play music according to their mood such as happy, sad, angry, or neutral. The program uses a graphical user interface (GUI) created with Tkinter and plays songs using the Pygame library.

2. Problem Statement

People often like to listen to music based on their mood. Searching songs manually takes time. So, this project aims to create an easy application where users can select their mood and instantly play music.

3. Technology Stack

- Programming Language: Python 3
- IDE: VS Code

4. Implementation

1. Create a dictionary of moods and songs.
2. Design GUI using Tkinter (buttons, labels).
3. When user clicks a mood button:
    - The corresponding song is loaded.
    - Music starts playing.
4. A stop button is used to stop the music.
5. Status label shows current activity.

5. Code Snippet

```
import pygame
import tkinter as tk

pygame.mixer.init()

songs = {
    "happy": "happy.mp3",
    "sad": "sad.mp3",
    "angry": "angry.mp3",
    "neutral": "neutral.mp3"
}

def play_music(mood):
    if mood in songs:
        pygame.mixer.music.load(songs[mood])
        pygame.mixer.music.play()
        status_label.config(text=f"Playing {mood} song", fg="green", bg="#000000")
    else:
        status_label.config(text="Invalid mood!", fg="red", bg="#000000")

def stop_music():
    pygame.mixer.music.stop()
    status_label.config(text="Music Stopped", fg="red", bg="#000000", font=("Wide
Latin", 45))

root = tk.Tk()
root.title("Mood Music Player")
root.geometry("500x500")

root.configure(bg="#000000")

title = tk.Label(root, text="Mood Music Player",
                font=("Algerian", 55),
                fg="yellow", bg="#000000")
title.pack(pady=10)
tk.Button(root, text="Happy", font=("Times New Roman", 30),
          width=50, bg="green", fg="white",
          command=lambda: play_music("happy")).pack(pady=20)

tk.Button(root, text="Sad", font=("Times New Roman", 30),
          width=50, bg="blue", fg="white",
          command=lambda: play_music("sad")).pack(pady=20)

tk.Button(root, text="Angry", font=("Times New Roman", 30),
          width=50, bg="orange", fg="black",
```

```
        command=lambda: play_music("angry")).pack(pady=20)

    tk.Button(root, text="Neutral", font=("Times New Roman", 30),
        width=50, bg="gray", fg="white",
        command=lambda: play_music("neutral")).pack(pady=20)

    tk.Button(root, text="Stop Music", font=("Arial", 30),
        width=20, bg="red", fg="white",
        command=stop_music).pack(pady=15)

    # Status label
    status_label = tk.Label(root, text="",
                    fg="white", bg="#000000",
                    font=("Arial", 12))
    status_label.pack()

    root.mainloop()
```

6. Sample Test Cases

Input (Mood)	Output
Happy	Plays happy.mp3
Sad	Plays sad.mp3
Angry	Plays angry.mp3
Neutral	Plays neutral.mp3
Invalid	Shows "Invalid mood"

7. Challenges Faced

- Handling missing audio files
- Managing GUI design properly
- Linking buttons with functions

8. Results and Observations

The program successfully plays songs based on the selected mood. It is simple, user-friendly, and works efficiently.

9. Future Improvements

- Add more moods and songs
- Add volume control
- Add playlist feature
- Detect mood automatically using AI

Project Submission

GitHub Link: https://github.com/TanaviVadagave/MoodMusicPlayer/blob/main/main.py

10. Signatures

Student Signature: _____

Guide Signature: _____

Date: _____